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A multidisciplinary investigation of the Notträsk Gabbro, northern Sweden

LENNART WIDENFALK, STEN-ÅKE ELMING and THOMAS ENMARK

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The differentiated, sulfide-bearing gabbro near the city of Boden in northern Sweden has been investigated in order to get a plausible model for its formation, shape and age. The gabbro is associated with a major fault system along which several intrusive bodies occur. The largest of these has been traversed by a gravimetric, magnetometric and geological profile. The investigated intrusion is undeformed and oval-shaped and covers some 20 km^2. The outer parts of the body consist of a norite in contact with older meatasediments and granites of varying age. Inside the norite occurs a magnetite-rich gabbro, which forms a ring around the centre of the massif, which in turn consists of leucogabbro–olivine gabbro. Fe, Ti and K are low in the centre where Mg and Cr have their highest contents. Impregnations of sulfides and a massive pyrrhotite vein with some 1 % Ni are located in the norite. A number of intrusion models have been tested by computer modelling combining gravimetric and magnetic field data. The most probable model seems to be a funnel-shaped body pointing downwards. The different gabbro types dip steeply inwards indicating vertical zones rather than horizontal. Palaeomagnetic measurements show that the intrusion has a similar orientation of magnetization as other gabbros of the Haparanda series in northern Sweden (1800–1900 Ga, Rb–Sr). Furthermore, the pole distributions show that the rock cooled slowly, which may indicate that the present outcrop is a deep part of an eroded intrusion. The massif is supposed to have been formed by a diapiric re-emplacement of primary horizontal cumulates formed in the sequence olivine gabbro–leucogabbro–magnetite-bearing gabbro–norite. □ *Gabbro, geology, geophysics, petrophysics, nickel, sulfide, magnetometry, gravity, palaeomagnetism, cumulate, Notträsk, northern Sweden, N6550, N6550, E2150, E2150.*

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A small area near the farm Notträsk, some 6 km NE the city of Boden in northern Sweden (see Fig. 1) was first claimed in 1890, when – according to old applications – high grade ore was found, but no mining took place. In 1975 the Notträsk gabbro became of interest again after the discovery of a massive pyrrhotite vein in a road cut. Since then several mining companies and the department of Economic Geology and Applied Geophysics at the University of Luleå have been involved in geological, geochemical and geophysical surveys in the area. Widenfalk (1979) presented the geology and geochemistry of the Notträsk intrusion in an unpublished report to the Geological Survey and some companies. This report was presented at the Geological Winter meeting in Bergen 1980. With Widenfalk as project-leader several Diploma-theses have been published (Sundin 1977; Svensson 1981; Börjeson & Nyberg 1981). The present work was originally a part of a Ph.D.-thesis (Enmark 1982) and was presented at the annual meeting of the Geological Society of Sweden in May 1982 (Elming et al. 1983).

Presently the LKAB Exploration Co is carrying out diamond drilling in the southern part of the area. Subsequent studies by the present authors have revealed that the Notträsk occurrence is located in a norite, which is a part of a regular and well differentiated gabbroic intrusion.

The formation of some nickel sulfide ores involves processes of sulfide–silicate immiscibility and magmatic differentiation and a far reaching differentiation of a gabbro magma may be of great importance for the ore formation. Since only minor tectonic events – faulting – have affected the Notträsk intrusion the internal relations are undisturbed. Hence, a more detailed study of this massif is justified in order to get clues to further ore prospecting in the area. Several small nickel occurrences have been found in this part of Sweden, from the Finnish border and further west and south-west (Grip 1961).

The two-dimensional geological picture obtained by traditional mapping showed an oval-shaped and differentiated intrusion that is so unusually well preserved that it was suspected of being of a very young age. Furthermore, the

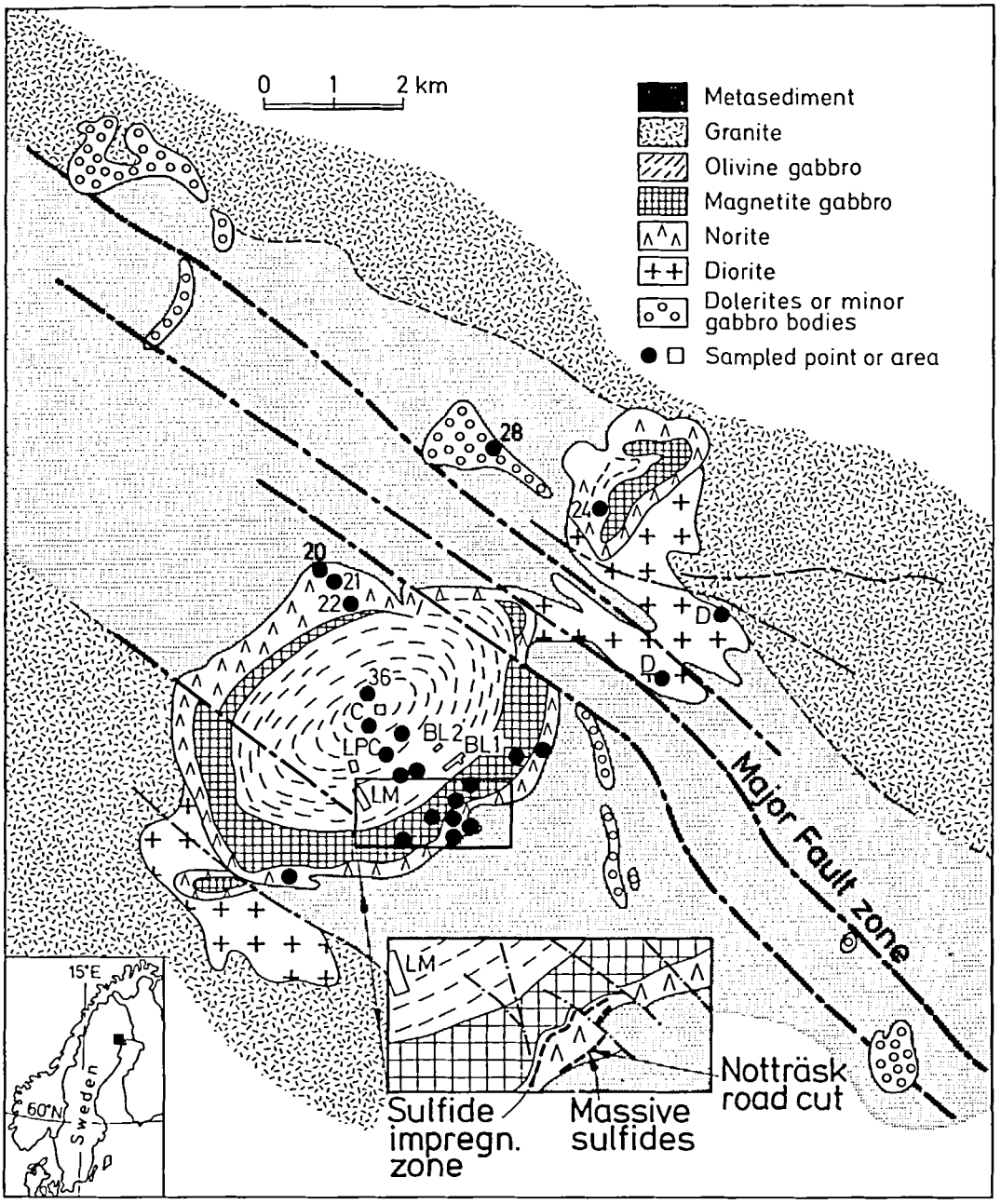


Fig. 1. Outlines of the geology of the Notträsk area.

major fault system running NW–SE through the area was also suspected of having some influence on the formation. Gravity and magnetic measurements have been carried out to test the plausibility of some models of the deeper structures. Palaeomagnetic investigations were undertaken in order to solve the problems of age. To restrict the number of possible geological models estimated from gravimetric and magnetic anomalies a good knowledge of densities and magnetic properties of the gabbro was deemed necessary. In order that a model should survive the plausibility test its calculated anomaly must agree with the measured data. It is not satisfactory that

elaborate geological models are often presented disregarding this geophysical constraint. However, it must not be forgotten that even if the calculated anomaly fits the measured data perfectly it is not necessarily the true model since we cannot get rid of the profound ambiguity in the interpretation of potential fields like gravity.

General geology

The outline of the local geology is shown in Fig. 1, which is based to a large extent on the aeromagnetic and ground surveys.

The geology of the area has been discussed by Lilljequist (1980) and the area can be described as a border zone between an Archean continent to the north in Sweden and Finland and Precambrian supracrustals to the south. The rocks have been extensively reshaped by repeated intrusions in several periods. The Notträsk gabbro belongs to the Haparanda series, a suite of differentiated and synkinematic intrusions in the folded Karelian supracrustals (Ödman 1957; Mikkola 1949; Härme 1949). In the investigated area the supracrustals mainly consist of metasediments. Ödman (1957) furthermore pointed out the similarities between Svecofennian and the Haparanda intrusives. This was verified by Welin et al. (1970, 1979) by Rb–Sr dating, who found the age of the rock series to be 1840 Ma, similar to that of synkinematic Svecofennian intrusives.

The rocks of the Haparanda intrusives vary from gabbro to granite and the main rock type is granodiorite. A number of gabbroic intrusions occur and several are nickel-sulfide-bearing (Grip 1961).

To the west the Notträsk gabbro is bordered by the younger (1550 Ma?, Lilljequist 1980) Edefors granite, which have had some, but limited, influence on the gabbro (see the discussion below).

Faults of different extension, mainly in NW–SE, cut the area (Lilljequist 1980). In connection with the major fault zone a number of small gabbro outcrops are seen.

Methods

The field work has been carried out in several periods from 1976 onwards. The first area to be surveyed was around the outcropping pyrrhotite vein (inset map Fig. 1). Here the number of outcrops is large making it easy to map and sample, but the contacts are mostly covered by moraine making contact relations difficult to determine. Therefore the distribution of the different rock types in this limited area (inset map Fig. 1) are mainly deduced from a magnetometric (vertical field) survey, made on a 80 × 20 m grid. A major part of the area has also been covered by an electromagnetic (slingram) survey (Sundin 1977). This survey delimited the extent of the sulfide occurrences.

The gabbroic rocks have been analysed at the Department of Economic Geology at the University of Luleå, using a plasma emission spectrograph.

Sampling for density estimations and in-situ determinations of susceptibility has been made mainly along a profile from the center to the outer part of the structure (Fig. 2). For a palaeomagnetic study the rock was sampled at 9 sites, 3 to 5 samples taken from each site, making in all 31 samples. These samples were drilled with a portable drill-machine and oriented with both sun and magnetic compasses. The specimens were measured in the non-magnetic laboratory with an astatic magnetometer and progressively demagnetized in alternating fields up to 55 mT. Density determinations were made on all specimens.

Regional gravity and low-altitude aeromagnetic maps have kindly been made available by the Geological Survey of Sweden (Figs. 3 and 4). Since the regional gravity stations are not evenly distributed and only a few are situated on the Notträsk gabbro a gravity and a ground magnetic (vertical field) profile across the massif was measured. The distance was 100 m between the gravity measurements and 50 m between the magnetic measurements. The location of the profile is shown in Fig. 2.

The gravity profile was tied to the regional field to get a 'zero level'. No terrain correction has been applied to the gravity measurements since the topography is relatively flat.

When computing the response from a model we have used a modified version of the computer program by Enmark (1981). The basic model in the program is a 2 1/2 dimensional body; i.e. a body of polygonal cross-section with the tails cut off in the strike direction. The program calculates the gravity or the magnetic field in an arbitrary point in the plane and is thus suitable for the profile across the gabbro.

Geology and petrography

The geological features of the investigated area are shown in Fig. 1 and the results of the chemi-

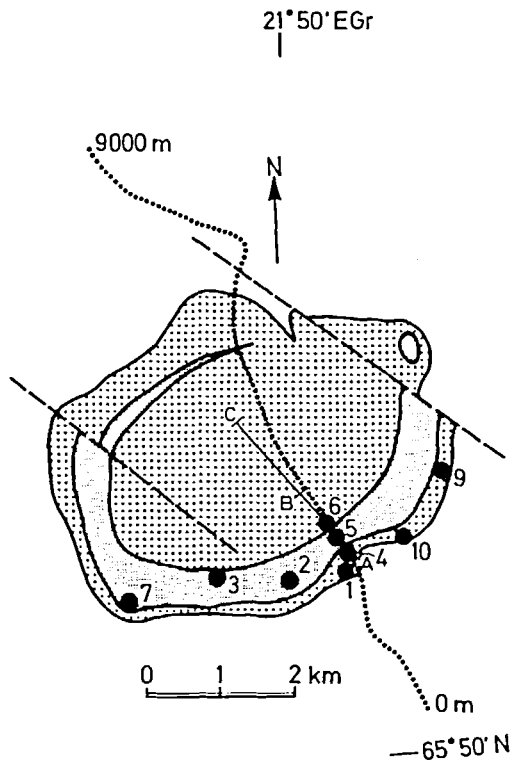


Fig. 2. Position of the palaeomagnetic sampling profile and distribution of the sites H1 to H10. Gravity and magnetic measurements were carried out along the dotted line, where every dot is a gravity and magnetic station. Additional magnetic mesaurements were done in between these stations.

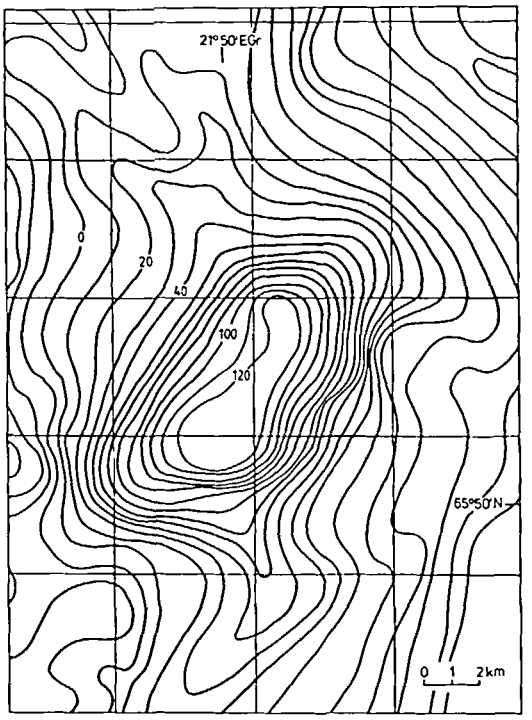


Fig. 3. Regional Bouguer anomaly map of the Nottråsk area. Contour interval is 10 g.u. Measuring density is 0.2 stations per km^2 on an average, but closer spaced in the central parts of the map.

cal analyses and microscopic work are summarized in Tables 1A, 1B and 2.

Besides the surrounding gneissic metasediments and granites the main rock types belonging to the mafic intrusion are: diorite, norite, magnetite-bearing olivine gabbro, leucogabbro, olivine gabbro and dolerite dikes.

Successively outwards from the centre of the gabbro body there is an increase of plagioclase, but diffuse layers show enrichment of clinopyroxene and olivine. Further out towards the borders the clinopyroxene content is raised and substantial amounts of biotite and oxides, magnetite and ilmenite, occur. As can be seen in the description of the various rock types there is a regular cryptic variation of the olivines and plagioclases. This was also shown by Arvanitidis (1982).

Although the topography in general is quite flat the various rock types may be recognized in field and on air photos because of variations in topography and elevation. This is probably due to different resistance to chemical and mechanical weathering. The most resistant rock is the magnetite-bearing olivine gabbro, which successively rises some 25 m above the surroundings.

Metasediments

Next to the gabbro body, both to the north and south, the gneissic metasediments show layering and are probably mafic tuffs. No primary textures can be recognized and the tuffs consist mainly of amphibole, plagioclase, quartz, microline, graphite and pyrrhotite. Because of their high density the tuffs could be delimited from the common metasediments by gravity measurements.

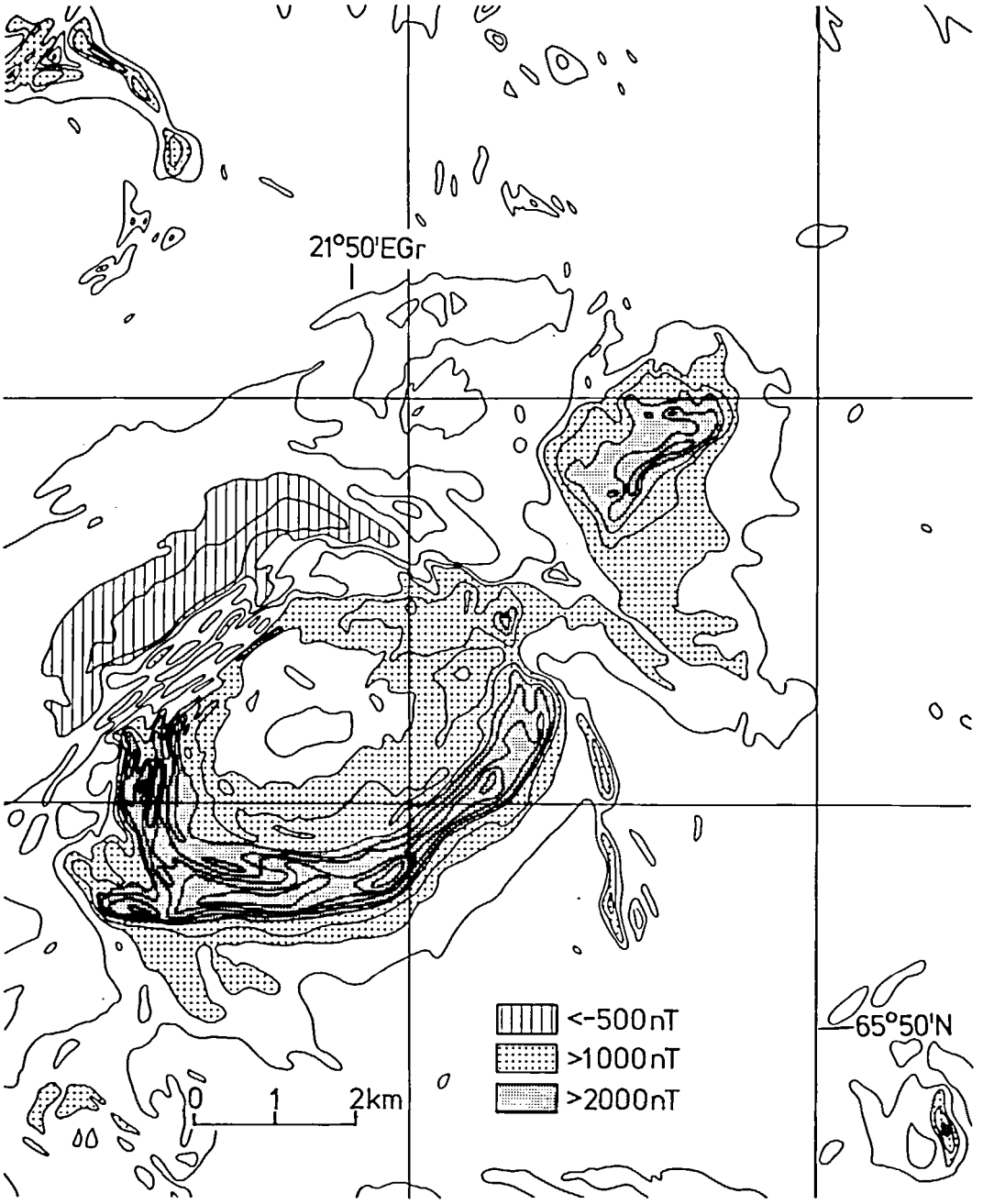


Fig. 4. Aeromagnetic map of the Notträsk area. Contour interval is 500 nT.

Diorite

Diorites outcrop to the southwest of the main massif and between the two largest intrusions within the fault zone. The texture is 'granitic' (hypidiomorphic granular) and it is rather even-grained with sizes from 2 to 5 mm. Oligoclase is the most frequent mineral and other minerals are in decreasing order of abundance: Clinopyroxene, K-feldspar, biotite, quartz and hornblende. Frequent myrmekites occur.

Table 1. □ A. Chemical compositions of the Notträsk intrusive rocks. For location of samples, see Fig. 1. □ B. CIPW norms of the analysed samples (in A). No Fe^{3+} determinations have been made but a constant Fe_2O_3/FeO ratio of 0.15 has been used.

Table 1 A

Sample	South margin norite	Magnetite-bearing gabbro	Central olivine-bearing gabbro BL1	LM	BL2	LPC	Centre	36	Northern margin norite 20	21	22	Satellite intrusion 24	28	Diorite	Dike
No of samples	6	6	42	45	9	5	5	1	1	1	1	1	1	2	1
Element (wt-%)															
SiO_2	51.2	50.5	49.1	49.3	48.1	49.1	47.3	44.0	51.7	49.5	50.7	50.7	49.9	59.3	47.0
TiO_2	1.16	1.55	0.29	0.35	0.22	0.27	0.26	0.17	1.21	0.97	0.67	0.68	3.53	0.73	1.93
Al_2O_3	15.2	16.0	21.3	20.4	21.7	19.7	18.5	14.6	17.6	16.6	17.9	20.9	14.8	16.3	16.0
FeO*	9.9	10.2	5.69	6.26	5.63	5.48	6.53	9.81	7.74	8.13	6.89	6.00	10.6	6.58	11.1
MnO	0.15	0.15	0.08	0.09	0.08	0.09	0.09	0.11	0.08	0.12	0.11	0.09	0.16	0.12	0.17
MgO	9.51	8.20	8.60	9.4	9.1	9.3	12.5	19.4	6.88	8.01	8.22	5.56	5.82	3.60	7.10
CaO	7.78	8.76	10.5	10.3	10.1	11.3	9.82	6.64	8.11	11.6	11.4	10.1	8.35	5.96	10.6
Na_2O	2.40	2.46	3.03	2.90	2.88	2.91	2.55	1.95	3.18	2.85	3.16	3.78	2.99	3.84	2.65
K_2O	0.99	0.92	0.38	0.39	0.34	0.33	0.31	0.23	0.84	0.63	0.63	0.84	1.45	1.80	1.56
P_2O_5	0.32	0.39	0.11	0.14	0.11	<0.05	<0.05	<0.5	0.34	0.25	0.05	0.55	0.39	0.25	0.69
Element (ppm)															
Ba	500	610	380	363	288	276	236	167	859	569	664	718	743	878	1000
Cr	330	300	NA	NA	NA	718	888	1420	377	193	566	49	29	56	240
Zr	80	58	NA	NA	NA	1	<1	<1	50	25	35	52	193	202	160
Cu	15	35	60	59	60	40	69	71	18	57	14	21	21	21	40
Ni	50	90	190	149	190	130	222	320	95	130	170	76	40	36	110
Co	25	53	65	65	71	67	84	110	53	67	71	67	67	47	50
Sum wt-%	98.7	99.2	99.2	99.6	98.3	98.6	98.0	97.2	97.8	98.8	99.9	99.3	98.1	98.6	99.0

Table 1 B

Sample	South margin norite	Magnetite bearing gabbro	Central olivine-bearing gabbros BL1	LM	BL2	LPC	Centre	36	Northern margin 20	21	22	Satellite intrusion 24	28	Diorite	Dike
No of samples	6	6	42	45	9	5	5	1	1	1	1	1	1	2	1
CIPW norm															
Q	–	–	–	–	–	–	–	–	0.3	–	–	–	0.2	9.9	–
C	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Or	5.9	5.4	2.2	2.3	2.0	2.0	1.8	1.3	5.0	3.7	3.7	4.9	8.6	10.6	9.2
Ab	20.3	20.8	25.6	24.5	24.4	23.9	21.2	16.4	26.9	24.1	26.4	32.0	25.3	32.4	19.5
An	27.8	29.9	43.3	41.5	45.2	39.7	38.1	30.4	31.2	30.6	32.7	37.5	22.6	21.9	27.1
Lc	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Ne	–	–	–	–	–	0.3	0.2	0.1	–	–	0.2	–	–	–	1.6
Di	7.1	9.1	6.5	7.0	3.4	13.2	8.5	2.1	5.5	20.6	19.0	7.3	13.3	5.0	17.2
Wo	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Hy	31.2	26.0	1.5	4.1	1.7	–	–	–	24.3	2.3	–	1.5	18.5	15.5	–
Ol	1.7	2.3	18.1	18.0	19.9	17.9	26.4	44.6	–	13.5	15.1	12.3	–	–	17.0
Mt	1.7	2.0	1.1	1.2	1.1	1.0	1.3	1.9	1.5	1.6	1.3	1.2	2.0	1.3	2.1
Il	2.2	2.9	0.6	0.7	0.4	0.5	0.5	0.3	2.3	1.8	1.3	1.3	6.7	1.5	3.7
Ap	0.7	0.8	0.3	0.3	0.2	–	–	–	0.7	0.6	0.1	1.2	0.8	0.6	1.5

Norite

Norite constitutes the rock type between the main massif and the gneiss or granites. No contacts or chilled margins have been found, but general field relations as well as the occurrence of gneiss fragments in the norite suggest that the mafic rocks intruded the metasediments. The marginal norite is well exposed in the road cut where the massive pyrrhotite vein occurs and this part of the gabbro contains also leucocratic veinlets besides gneiss xenoliths.

The average norite is evengrained, around 2 to 3 mm, but varies up to some 6 mm. The modal composition (Svensson 1981 and this investigation) is shown in Table 2.

Most minerals are fresh but the plagioclase, varying from oligoclase to andesine, is somewhat

Table 2. Modal composition of selected samples from the main rock types of the Nottträsk intrusion.

Sample/ Mineral	Southern norite 13	Magnetite gabbro 3	Leuco-gabbro BL 2	Olivine gabbro 36	Northern norite 21	Dike 7
Quartz	1	0	0	0	0	0
Plagioclase	56	65	79	57	53	55
Olivine	0	10	4	35	0	6
Orthopyroxene	20	0	0	0	21	1
Clinopyroxene	5	15	2	6	11	11
Amphibole	1	1	0	0	1	0
Biotite	5	4	1	0	8	17
Accessory and ore min.	12	5	4	2	6	10

sericitized near the massive sulfide vein, where also some quartz occurs.

The pyroxenes are sometimes altered to chlorite. The orthopyroxene content varies and a major part of the marginal norite should be termed gabbronorite (IUGS 1973). Some clinopyroxene occurs and is partly symplectic intergrown with plagioclase. Microcline with perthite-lamellae are present as interstitial grains and when in contact with plagioclase well developed myrmekites occur. Small euhedral grains of apatite are common. Biotite is an interstitial mineral and appears together with ilmenite and other ore minerals. Olivine, Fo 70, has been found in a few samples (Svensson 1981).

The texture of the minerals is generally hypidiomorphic, but cumulus plagioclase with poikilitic clinopyroxene is not uncommon. Sample 24 from the largest satellite intrusion shows well developed cumulus orthopyroxene.

Ore minerals are frequent, generally with an anhedral habit. Ilmenite is the most common ore mineral, but near the upper and lower norite borders pyrrhotite may dominate. Also magnetite, chalcopyrite and pentlandite have been observed.

Sulfide occurrences

The electromagnetic survey revealed that the marginal norite contains two different sets of conductors, both caused by impregnations of sulfides, mainly pyrrhotite. The Ni- and Cu-richer impregnations and the massive pyrrhotite lens occur in the outermost part of the norite near the gneiss. The other set of sulfide impregnations, the inner sulfide zone, appears quite near and parallel to the magnetite-bearing gabbro.

The thickness of the massive lens is only two or three meters and with an impregnated zone of about 30 m surrounding it. According to the geophysical survey the length of the vein may be some 100 m, but the depth extent is unknown. The borders of the massive vein are very irregular. The number of inclusions increase outwards and the lens turns within a meter into an impregnated norite.

The massive pyrrhotite vein is built up mainly of pyrrhotite with some chalcopyrite and pentlandite. The pyrrhotite is rather coarse grained with grain sizes up to 10 mm. The chalcopyrite, on the other hand, is more fine-grained (0.1–0.3 mm) and it occurs mostly interstitially between the larger pyrrhotite grains. In some cases chalcopyrite is somewhat enriched in the outer parts of the massive lens at the contacts with the norite.

Two different types of pentlandite appear; mainly separate grains but also tiny lamellae (<0.1 mm) within larger pyrrhotite grains. As a rule the separate pentlandite grains are elongated and up to 2 mm in length with frequent cracks. Along these cracks it is often altered to violarite.

Accessory minerals are gersdorffite, niccolite, sphalerite, ilmenite and magnetite. Also some graphite occurs possibly indicating contamination from the footwall metasediments. Electron-microscopic studies have been made on pyrrhotite from the inner sulfide zone and pyrrhotite from the massive ore. It was found that the latter contained small amounts of Co and Ni, while these were not detectable in the pyrrhotite from the inner zone. These studies also showed that the Co-content is mainly bound to the larger pentlandite grains, while the small pentlandite lamellae within the pyrrhotite contained no detectable amounts of Co. Arvanitidis & Rickard (1979, 1980) investigated the pentlandite–pyrrhotite relations and found that the mineralogy indicated a far reaching exsolution caused by a very slow cooling of the rocks.

The average of six analysed, massive ore samples show 1.05 % Ni, 0.21 % Cu and 0.16 % Co. Recent drilling by the LKAB Exploration Co (Lars-Göran Ohlson, Stockholm, 1982, personal communication) has revealed a nearby massive lens with approximately 0.8 % Cu, 0.8 % Ni and 0.2 % Co.

Magnetite-bearing olivine gabbro

The main mineralogical differences between the marginal norite and the magnetite gabbro are the increased contents of magnetite, olivine and clinopyroxene in the latter. Olivine and clinopyroxene appear in about equal proportions and orthopyroxene occurs only occasionally. The composition of the olivine is Fo 60–65 and the plagioclase is andesine–labradorite. The plagioclase laths are up to 6 mm long and sometimes show signs of strain. They are occasionally faintly zoned. The clinopyroxene appears to a great extent as poikiloblasts (< 10 mm) and most samples show cumulate textures. Biotite as well as the ore minerals are interstitial and commonly less than 2 mm in size. The dominating ore mineral is magnetite, up to 7 % by volume, and ilmenite (Sundin 1977). This content may be too low for the rock to be called magnetite gabbro, but the contrast with the surrounding gabbro types is so pronounced that the term may be justified as a 'working term'. The maximum magnetite contents are near the outer border towards the norite. The major part of the Ti is localized in anhedral ilmenite and only a small fraction appears as ilmenite-lamellae in magnetite.

The overall sulfide content is low but pyrrhotite occurs in most samples as small, well rounded droplets. These are often made up of one or two pyrrhotite grains and sometimes with a small chalcopyrite grain at the outer contact.

The magnetite-bearing olivine gabbro has similarities with the titaniferous iron ore deposits in some differentiated Swedish gabbros; e.g. the Rackeby norite (Stålhös 1958) and the Kramsta gabbro suite (Lundegårdh 1957).

Leucogabbro and olivine-bearing gabbro

The composition of the inner part of the intrusion differs significantly from the outer parts. The contents of Mg, Na, Ca and Al are generally higher whereas the Fe, Ti and K contents are much lower than the surrounding gabbro. The contents of olivine (Fo 75–82) and labradorite are increased and the biotite is much lowered. Clinopyroxene vanishes to a great extent and occurs as poikiloblasts. Orthopyroxene appears rarely interstitially and as rims on clinopyroxene and olivine. Chromite may be seen as small rounded grains and often as inclusions in the olivine.

The great number of analyses of the olivine gabbro and leucogabbro (BL 1, BL 2, LM, LPC in Table 1A–B) is due to the use of those rock types in blast furnaces in iron-works because of suitable chemical composition.

Plagioclase and olivine of the inner gabbros are in general euhedral and the rocks are considered to be cumulates.

As can be seen in Table 1A–B there is a variation in the inner gabbros, and three main tendencies may be identified: enrichment of plagioclase (BL 2 samples), enrichment of olivine (centre samples) and enrichment of clinopyroxene (LPC samples). This is illustrated in the triangular diagrams, Figs. 5 and 6. The samples mentioned above are taken at intervals of 10 m and in some cases the Al_2O_3 content may vary with 3 to 5 % from sample to sample. The sample with a low Al_2O_3 content is an olivine gabbro and the higher content indicates a leucogabbro. This variation is as a rule gradual and rythmic layering like in layered intrusions is seen only occasionally. The strike is then parallel to the borders of the body. In the centre of the intrusion (sample 36) the SiO_2 content is only 44 wt-% which may indicate a more or less hidden ultramafic core. The innermost parts of the centre of the intrusion are troctolite.

Dikes

The main fault direction in the investigated area is NW–SE (Fig. 1). which also coincides with direction of most dikes. Furthermore, near the massive pyrrhotite lens, it was seen that a dike is cutting a sulfide-rich impregnation zone, thus showing that the dikes are later than the mineralizations. Where the dikes and faults cut the norite some mineralized norite-breccia, probably a result of local redistribution of the sulfides, has been found.

Only one dike has been analysed (sample 7, Table 1). Compared to the other analysed rocks the high contents of Ti and K seem to be the most outstanding feature. The minerals are ophitic andesine, olivine (Fo 53), clinopyroxene and biotite. Accessory minerals are chlorite, ilmenite and magnetite. The average grain-size is less than 0.5 mm. Near the outer contacts the plagioclase laths are oriented parallel to the border.

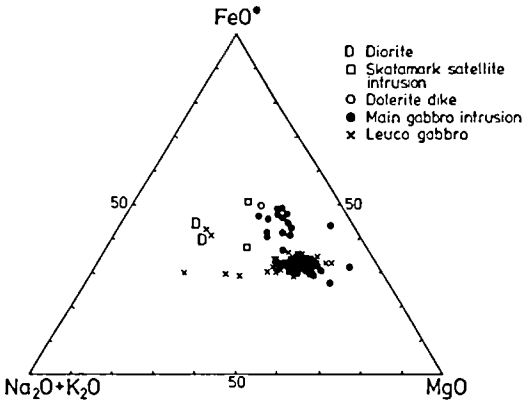


Fig. 5. AFM diagram showing the calc-alkaline trend for the Notträsk gabbro.

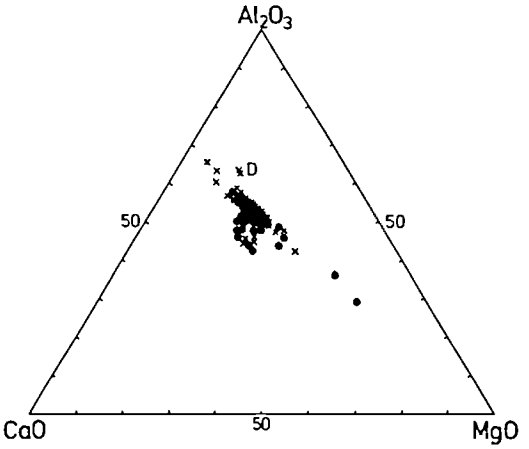


Fig. 6. CaO–MgO–Al_2O_3 triangular plot for the Notträsk gabbro. For explanation of symbols see Fig. 5.

Density and magnetic properties

The compositional differentiation mentioned is well reflected in the density variations. Although the number of samples are few, the high-magnetic zone recognized from the aeromagnetic measurements is quite well defined by the relatively high densities (2930–3000 kg/m^3; Fig. 7A) which can be partly explained by the increase of magnetite content. Samples from the marginal norite show large variations in density. Some of them are rich in pyrrhotite, but generally the densities are similar or slightly lower (2950 kg/m^3) than the densities of the high-magnetic zone (2960 kg/m^3). Crossing the zone the density decreases (2800–2850 kg/m^3) to increase to 2900–3000 kg/m^3 in the central parts reflecting the accumulation of olivine. The metasediments partly surrounding the formations have a higher density (2970 kg/m^3) in contact with the gabbro than elsewhere.

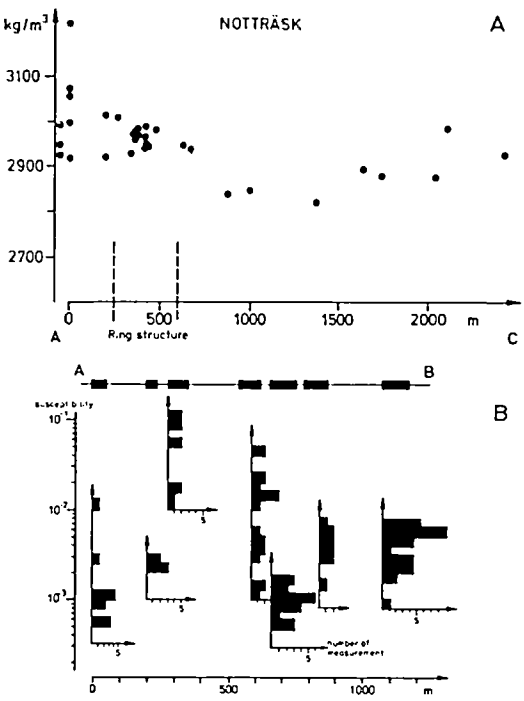


Fig. 7. □A. Density and □B. susceptibility variations estimated along the profile in Fig. 2.

The strongly magnetized ring structure is well recognized on the profile of in-situ measurements of susceptibility (Fig. 7B). The susceptibilities are measured on outcrops, the extents of which are shown along the line AB in Figure 7B, from the norite at the margin to the inner part of the structure. Outside the magnetite-rich zone the susceptibility of the noritic rock is low (~0.001). In the high-magnetic zone the susceptibility varies but is definitely much higher (0.001–0.1). At the inner contact with the ring the susceptibility decreases to the same level as outside the zone, to increase again (0.001–0.01) towards the central part.

An empirical relation between density and susceptibility of basic rocks from the north of Sweden has been established by Henkel (1976). This relation is expressed by the dashed lines in Fig. 8 and makes a reduction of the volume content magnetite from the density possible giving silicate densities of the rocks. In the density–susceptibility diagram (Fig. 8) the specimens of the palaeomagnetic study have silicate densities corresponding to a gabbroic composition. Only in

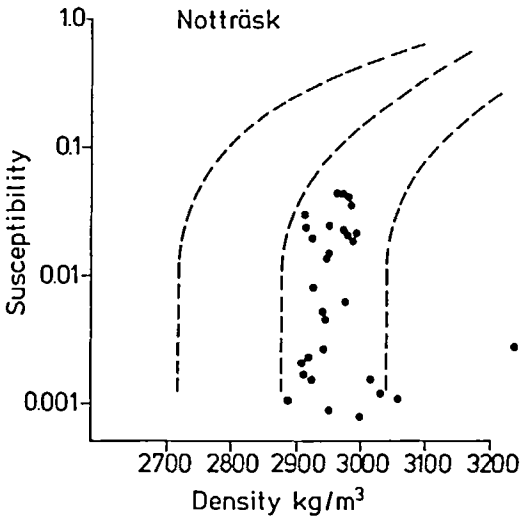


Fig. 8. Distribution of the density and susceptibility. Dashed lines indicate the empirical relations between density and susceptibility of mafic rocks from northern Sweden (Henkel 1976).

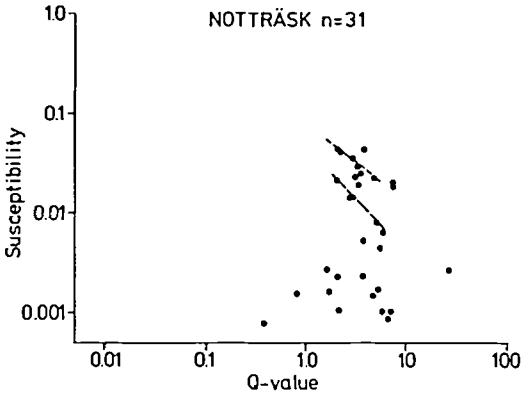


Fig. 9. Distribution of Q-value and susceptibility. Dashed lines show the within site relation between Q-value and susceptibility.

the ringstructure the content of magnetite is high enough to affect the density. The susceptibility of the samples never exceeds 0.05 and the maximum in-situ measurements is 0.1. The Q-value susceptibility diagram (Fig. 9) shows high Q-values, in general between 1.5 and 8.0. The susceptibility varies in the gabbro body but is locally rather homogeneous. Within some of the sites a trend of increasing Q-values with decreasing susceptibility is noticed. This is probably explained by grain size variations in the magnetic minerals influencing the magnitude of susceptibility and remanence in opposite ways (Parry 1965; Elming 1982*a*). No systematic difference in the Q-values from different parts of the gabbro intrusion is noticeable, which probably means that the grain size variations and/or crystallographic defects in the magnetic minerals are about the same throughout the sampled area.

When the soft part of the remanence is removed the characteristic remanence is in general relatively hard (Figs. 10B and 11B). The contribution of soft remanence to the total remanence varies from quite small in the sites H2, H3, H5, H6 and H9 to large in the site H10.

Samples from site H10 have quite high susceptibilities and intensities of remanence considering that they come from the outer part. The amount of sulphides in this noritic rock is high and pyrrhotite probably forms an important part of the magnetic minerals. Two of the samples of site H1 have a high percentage of pyrrhotite which is also reflected in the densities 3056 and 3218 kg/m^3. The high amount of pyrrhotite is not accompanied by any appreciable increase in susceptibility which indicates that the hexagonal type of pyrrhotite is dominating. This is supported by the study of pyrrhotite from this area made by Arvanitidis & Rickard (1980) showing that the monocline pyrrhotite constituents only 5 to 10 % of the total amount of pyrrhotite. The remanence of the samples at this site is generally soft. However, one sample rich in pyrrhotite and ilmenite, has a characteristic remanence which is very hard and quite stable. The remanence of pyrrhotite is usually soft but in basic rocks pyrrhotite with very hard remanence has been noticed (Elming 1982*b*). Without making a thermal demagnetization of this specimen it is thus impossible to say whether the carrier of the remanence is pyrrhotite or ilmenite or both.

The samples from site H7 situated in the margin of the ring structure (Fig. 2) show somewhat different magnetic properties. The remanence is to a great extent soft (~ 50 %) and the characteristic remanence decreases faster than for the majority of the samples (Fig. 11B).

Direction of remanent magnetization

The directions of NRM (Natural Remanent Magnetization) are very scattered and form a poorly defined mean of 349° decl. and 43° incl. (Fig. 12). Already after demagnetizations in 7.5 to 15 mT AF (Alternating Field), the directions of remanence are generally stable. Changes in directions from NRM, seen mainly after the first steps in demagnetization, from generally higher inclinations towards lower ones indicate the secondary

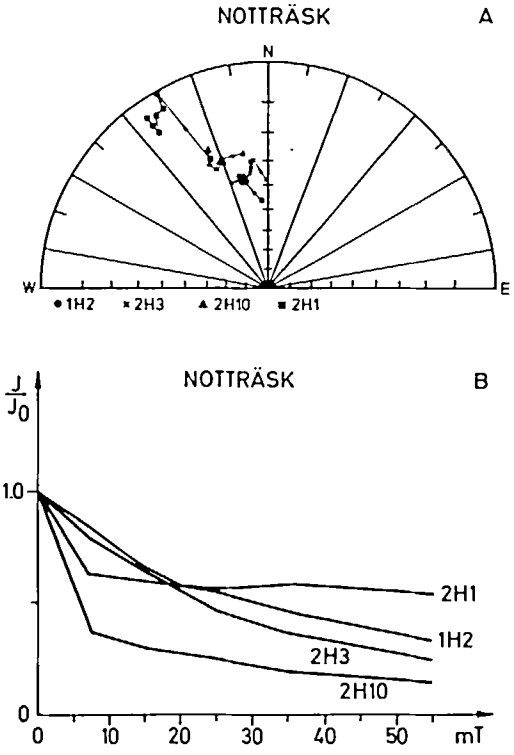


Fig. 10. Behaviour of □A. the directions and □B. the decay of the remanent magnetization with progressive AF.

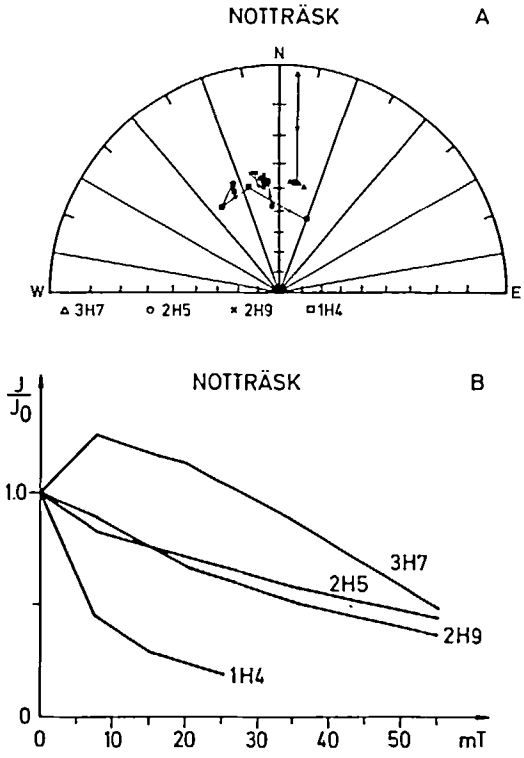


Fig. 11. Behaviour of □A. the directions and □B. the decay of the remanent magnetization with progressive AF.

soft component to be influenced by the Earth's present magnetic field. After the specimens are demagnetized in 20 mT AF the remanence decreases at an almost constant rate and is very stable (Fig. 10B), which probably means that only the characteristic remanence is left at this stage. The direction of the characteristic remanence from the specimens of site H7 is stable (Figs. 10A and 11A), but since it is significantly different from the majority of site (Fig. 13, Table 3) it is not used for the mean for the gabbro body. The difference in direction for site H7 cannot be explained by local tilting since the observed fault movements are small and in another direction. Also the mean directions of site H1, poorly defined from two samples, is well separated from the rest of the site means. This site mean is also disregarded for estimating the formation mean. The samples from the three sites H4, H5 and H6 situated outside, within and inside the ring structure are all sampled within a distance of about 700 meters. As their directions of remanence are similar they are all treated together as one site. Excluding the sites H1 and H7 and treating the sites H4, H5 and H6 together the sites used for the formation mean form a mean with a high precision at 25 mT AF (346,2° decl. 35.0° incl.; Table 3).

Apparent pole positions

Supposing the blocking temperature range of the individual sites is wide enough to average secular variations, the site means can be regarded as palaeomagnetic poles.

The site mean poles used for the formation mean are grouped quite close together giving a formation pole position at 219.7°E 42.6°N (Fig. 14, Table 3).

The pole of site H7 is at about the same latitude but 20 degrees further west. The rock at this site, situated in the margin of the gabbro, is metamorphosed: biotite and pyroxene altered to chlorite, ilmenite and some magnetite and remagnetized during the process so that this pole represents a magnetization age younger than the poles determined from other sites in the gabbro

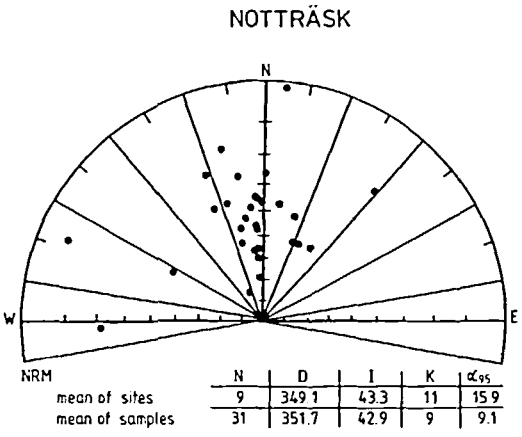


Fig. 12. Directions of the Natural Remanent Magnetization (NRM) of the specimens from the Notträsk gabbro.

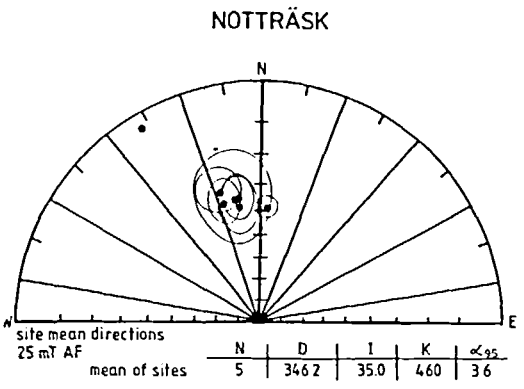


Fig. 13. Directions of the site means of the remanent magnetization with cone of confidence (95 % probability level).

do. The poorly defined site mean of site H1 gives a pole position well separated from the others. This site is situated outside the strongly magnetized ring, close to the border of the gabbro. In this area the intrusions of basic dikes, intruded in the final stage of the rock formation, have lead to local alterations, redistribution of the sulfides and also to brecciation of the rock. These magmatic-metamorphic activities most probably explain the anomalous pole position of the site H1.

Geological mapping shows some fault zones crossing the gabbro body. The magnetic anomalies indicate that the gabbro, if intruded in a symmetrical shape, is somewhat tilted from its original position (see discussion below). If the tilting has occurred after the rock has been magnetized the mean pole of the formation should be moved from 42.6°N and 219.7°E to 45.9°N and 217.2°E. This difference in the mean pole position is almost within the error of pole determination.

Gravimetric and magnetic investigations

The Notträsk gabbro stands out extremely well on both the aeromagnetic and the gravimetric maps (Figs. 3 and 4). An almost perfectly ellipse-shaped structure can be seen on the magnetic map. The structure consists of a high-magnetic (>2000 nT) ring surrounding a relatively low-magnetic (<1000 nT) central part. The high-magnetic zone, built up of several thin zones, is

Table 3. Mean directions of the sites and the whole gabbro complex with associated apparent pole positions. 25 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole °E	Palaeomagnetic pole °N	dp	dm
H1	2	318.4	−3.1			245.6	16.3		
H2	5	348.9	34.9	82	8.5	216.2	42.8	5.6	9.8
H3	2	348.1	35.1			217.2	42.9		
H4, 5, 6	7	349.1	38.2	31	11.0	216.2	45.1	7.7	13.0
H7	3	3.1	39.3	792	4.4	197.7	46.3	3.1	5.2
H9	5	342.9	35.7	83	8.4	223.9	42.5	5.7	9.8
H10	3	342.2	31.1	195	8.9	224.1	39.5	5.5	9.9
Mean of sites 2,3,5,9,10	5	346.2	35.0	460	3.6	219.7	42.6	2.4	4.1
Mean of samples	22	346.6	35.7	61	4.0	219.3	43.0	2.7	4.6

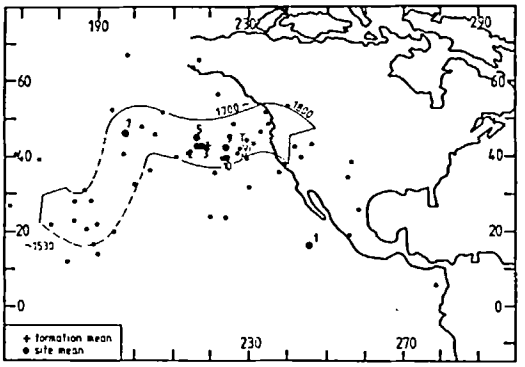


Fig. 14. Apparent pole position of the formation and site means in this study and other apparent poles (small dots) from the Baltic shield (Piper 1980; Pesonen & Neuvonen 1981; Elming 1982*a*). The solid swathe indicates the apparent polar wandering path for the time sequence 1800–1700 to 1530 Ma (Elming 1982*a*). Vi = Vittangi gabbro, T = Tärendö gabbro, Ni = Niemisel gabbro, all in northern Sweden.

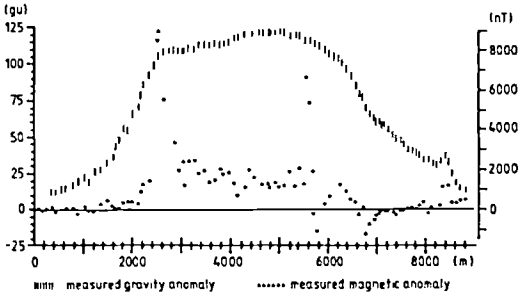


Fig. 15. Projection of the gravity and magnetic profiles seen from NE.

cut by a number of faults and even disappears in the NE part of the structure. A good correlation between the in-situ susceptibility measurements (Fig. 7B) and the magnetic pattern of the massif is seen. Furthermore from the magnetic measurements it appears that the magnetic zones must be steeply dipping but pointing inwards, i.e. towards the centre.

There are at least four satellites to this gabbro, one of them considerably bigger than the others, outside the main body. All the satellites line up along the major fault zone striking SE–NW (Fig. 1). The magnetic map shown in Fig. 4 is a simplified plot of the original map and thus some details may have disappeared. However, the original aeromagnetic map shows that the fault direction is mainly SE–NW and only a few minor indistinct faults can be detected in other directions.

The gravity map of the massif shows a positive anomaly of about 120 gravity units (equal to 12 mgal), Fig. 3. This anomaly is more elongated than the magnetic one. Probably the elongation is caused by the satellite body NE of the main intrusion having its gravity centre close to the fault zone and west of its magnetic centre, indicating that the Major Fault Zone (Fig. 1) was trigging the genesis of the whole massif.

Comparing the gravity and magnetic profiles (Fig. 15), it is seen that the locations of the maximum gradients of the anomalies, which often indicate contacts between different rock types, are somewhat offset. This is most probably caused by variations in density and susceptibility of the metasediments. Geological and petrophysical studies support the idea that a mafic tuff partly surrounds the Notträsk gabbro. This tuff is a part of but heavier than the metasediments.

Discussion

Palaeomagnetism

As the uplift rates of pre – 1000 Ma mobile zones (Watson 1976) were slow, the east–west distribution of the apparent poles in this study and other palaeomagnetic studies of basic rocks in northern Sweden (Piper 1980; Elming 1982*a*) might at least partly be explained by a slow rate of cooling due to uplift. A long period of magnetization is supported by the orogenesis, which from the time of the primorogene intrusions (represented by the Haparanda suite ~ 1840 Ma) to the postorogene intrusions (the Edefors granite?) represents an evolution over at least 250 Ma. Indications of slow cooling of this special formation also exists. In the dioritic and noritic parts myrmekites frequently occurs and Arvanitidis & Rickard (1979, 1980) found in their investigation of the pentlandite pyrrhotite relations, that the mineralogy indicated exsolution caused by very slow cooling of the rocks.

It is hard to estimate the influence of the intrusion of the serorogene granites and the Edefors granite on the magnetization of the rock. However, the metamorphosed rock at site H7 is definitely remagnetized and being close to the Edefors granite it has probably been affected by its intrusion.

The formation mean pole position is somewhat more westerly than but near the apparent poles of the Niemisel (Piper 1980), the Vittangi (Elming 1982*a*) and the Tärendö (Cornwell 1968;

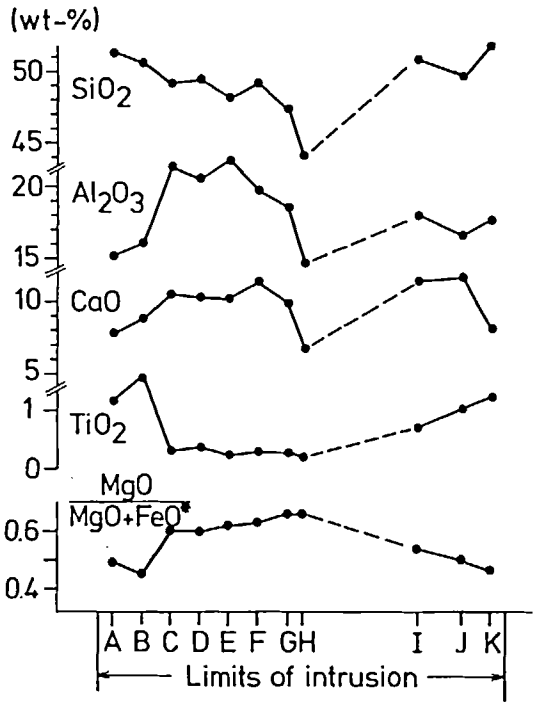


Fig. 16. Showing the schematic profile across the Nottträsk gabbro. Variations of some major elements. The letter symbols: A = southern norite, B = magnetite-bearing gabbro, C = sampled area BL1, D = sampled area LM, E = sampled area BL2, F = sampled area LPC, G = sampled area centre, H = sample 36, I = sample 22, J = sample 21, K = sample 20.

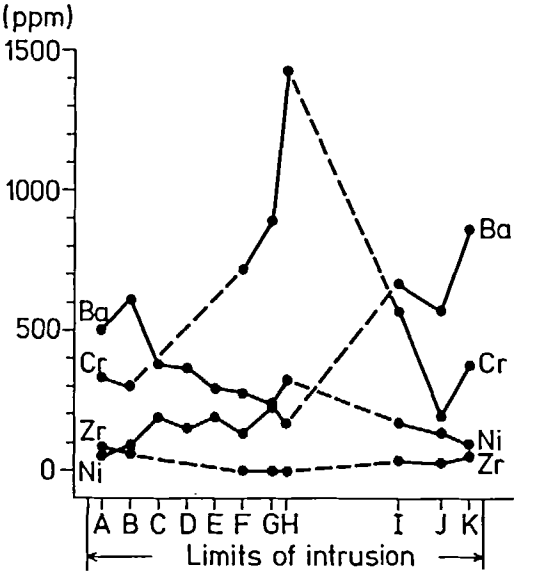


Fig. 17. Schematic profile across the Nottträsk gabbro showing the variation of some minor elements. For letter symbols, see Fig. 16.

Elming 1982*a*) gabbros, all of which most probably belong to the Haparanda suite. The site means all fall within the swathe indicating the apparent polar wandering path from the Baltic shield for the time sequence ~1800~1700 to 1530 Ma (Elming 1982*a*; Fig. 14 herein).

The slow uplift and a long metamorphic sequence have probably lowered the cooling rate and thereby also the blocking temperatures of the magnetic minerals (Pullaiah et al. 1975). This makes a correlation of the age of magnetization to the radiometric ages of the rocks in the area difficult.

The formation mean pole position represents an age of magnetization younger than 1840 Ma and probably somewhat younger than the serorogene intrusions (1770 Ma; Elming 1982*a*). Using the apparent polar wandering rate 0.35°/Ma for the Baltic shield (Pesonen & Neuvonen 1981) the metamorphic rock at site 7 seems to be remagnetized 55 to 75 Ma later than the main part of the gabbro intrusion.

As the Q-values of the magnetic minerals in this formation are generally high (1.8 to 8.0), a knowledge of the magnitude and direction of the remanence is absolutely necessary for the interpretation of the magnetic data. Furthermore, the variations in intensities of magnetizations and densities, caused by the differentiation of the rock, also make a subdivision of a possible model necessary.

Geology

The AFM diagram (Fig. 5) shows that the Nottträsk intrusion falls within the calc-alkaline field (Irvine & Bargar 1971), but quite close to the subdivision line to the tholeiite field.

The chemical analyses as well as the varying mineralogy show that the massif is well differentiated from an ultramafic core to a partly quartz-bearing norite in the outer parts. The variation of SiO_2 is also accompanied by a regular change of the MgO/(MgO + FeO*) ratio and e.g. the Ni and Cr contents, which are all high in the centre (Figs. 16 and 17). This should be expected if they constitute an early cumulate fraction of a mafic magma.

The intrusion margin, consisting of norite and gabbronorite has high amounts of SiO_2 and Zr and very low contents of the ore forming elements Ni, Cu and Co, a fact that may contradict the appearance of sulfide veins and impregnations.

The satellites seem to be enriched in TiO_2 and/

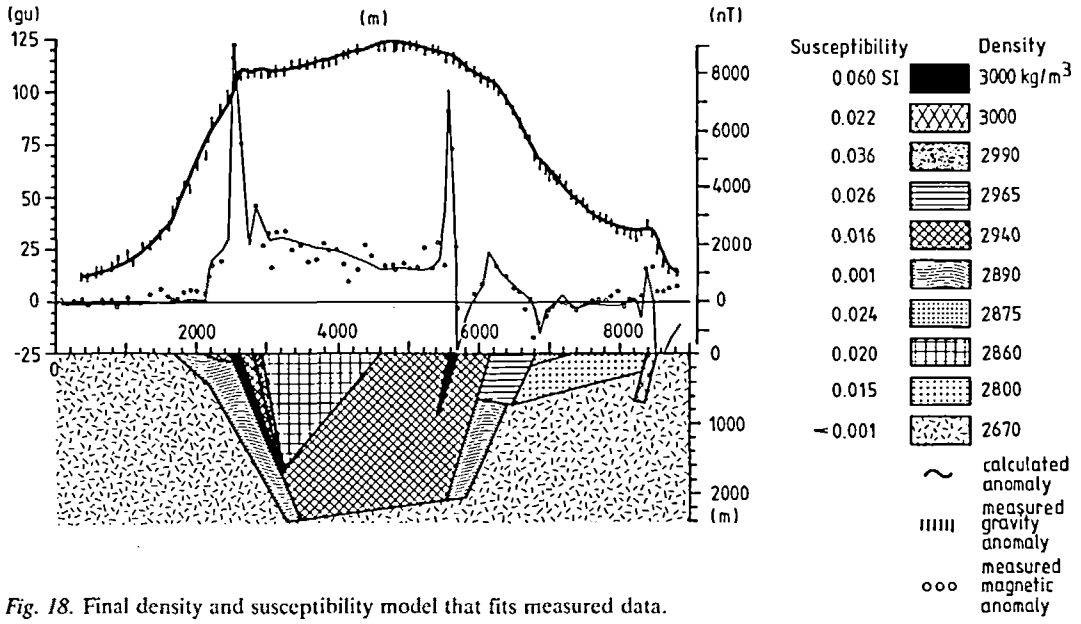


Fig. 18. Final density and susceptibility model that fits measured data.

or alkalies, which also increase outwards in the main intrusion.

Taking into account all the observations discussed above, we propose a model for the structure and origin of the Notträsk gabbro according to which it is a mafic intrusion made up of different cumulates during successive magmatic differentiation.

The data fit an evolutionary model with the following rock sequence, from early cumulates to the late dikes: (ultramafics) – olivine gabbro – leucogabbro – magnetite-bearing olivine gabbro – norite – sulfide mineralizations – dikes (-diorite). A rock series that seems to be identical to that of the gabbroic rocks of some Alaskan-type ultramafics (Irvine 1974). There, as well as in Notträsk, the late appearance of oxides is a result of increased oxygen fugacity due to higher water content in the remaining magma. Hydrous minerals also increase outwards in the Notträsk intrusion. The geophysical investigation discussed below indicates that the layering is vertical although the dominating rock fractions are cumulates. Rythmic layering as shown by Wager & Brown (1967) is not frequent, but when it occurs, the dip and strike coincide with our model.

Gravity and magnetics

If all geological, petrophysical and geophysical data are combined a number of models of the Notträsk gabbro must be rejected. One fundamental constraint on an acceptable model is that it must fit the observed geology and the petrophysical data. If the observed densities in the massif and its surroundings are representative also of a deeper section, gravity data limits the depth of the structure to approximately 2 kilometres. Probably this is a maximum depth since the densities most likely increase downwards (Henkel 1981).

That a model must fit both the gravity and the magnetic data turns out to be a severe restriction and only a few models can pass this test. Using the data discussed above as a starting attempt a final model fitting both gravity and magnetic measurements (Figs. 18 and 19) was reached after some small modifications. According to the final model the intrusion is somewhat tilted, as mentioned in the previous section, which may be indicated by the faults shown in Fig. 1. The zoning seen both geologically and petrophysically is also included in the model and shown in Fig. 19 as a plan view. Even if this is a density/susceptibility model it may be directly compared with the geological map Fig. 1. Looking at the calculated anomalies it can be seen that these data show an almost perfect fit to the measured gravity field while differences exist in the magnetic profile. These differences are probably due to near surface magnetic rocks and/or blocks.

Although this model fits the presently available data satisfactorily, it can never be proved to be the true one, because it may be refuted in the

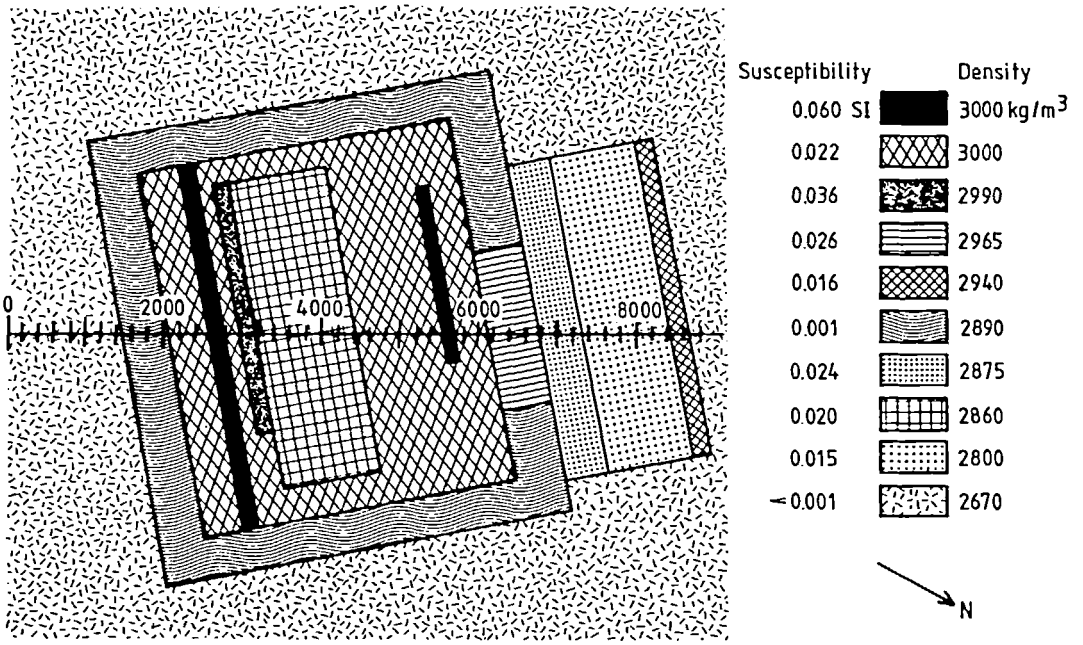


Fig. 19. Plan view of the model in Fig. 19. The section is shown at 50 metres depth.

light of other considerations and data. Keeping the model as a whole undisturbed it is found, like in the paper by Enmark & Nisca (1982), that only small changes in some selected parameters can be allowed. For example, the slope of the contact between the metasediments and the gabbro can only be changed some degrees before the difference between measured and calculated values becomes clearly visible. Another similar example is the depth extent of the model. Keeping all other parameters except the depth extent of the gabbro fixed, it is easy to show that changing the depth extent of the gabbro even a few hundred metres is not possible.

Many other parameters can be studied in the same manner, if desired and called for during subsequent investigations of the structure such as drilling.

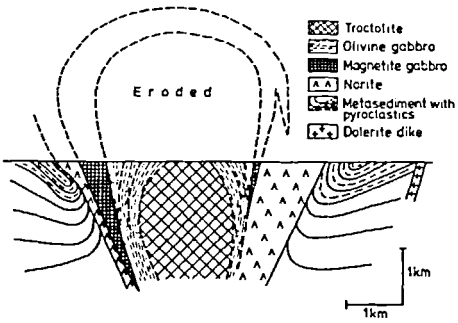


Fig. 20. Proposed structure model for the Notträsk intrusion.

Conclusions

In an attempt to explain the differentiation in Notträsk, Widenfalk (1979) pointed out the possibilities of flowage differentiation (Bhattacharji & Smith 1964; Bhattacharji 1966). This mechanism was also favoured by Arvanitidis (1982) as an explanation for the differentiation of the Notträsk gabbro. However, this differentiation process requires a rather narrow feeder channel and could be disregarded in geophysical model fitting calculations because of its negligible anomaly. Hundreds of gabbro intrusions occur in northern Sweden and some have geophysical patterns similar to that of Notträsk. Henkel (1981) proposed that they were intruded at a great depth and with a magmatic differentiation similar to that now established for the Notträsk intrusion. However, the vertical zoning must be explained and for this the mode of intrusion of the Alaskan ultramafic massifs (Irvine 1974) was used as a base for our model.

The Alaskan ultramafics intruded at great depth within the crust and cooled slowly. After the solidification or in a semi-solid state the

deepest rocks – dunites – continued to rise as diapirs and intruded the later differentiates. The proposed model for the Notträsk intrusion is shown as a profile across the massif (Fig. 20).

The existence of two types of sulfide mineralizations, one outer with substantial amounts of Ni, Cu and Co and one inner sulfide zone poor in these elements may contradict the fact that the norite has the least amounts of these elements of the investigated rocks. The explanation of this phenomenon may be separate formation of parts of the mineralization and the norite. According to Duke & Naldrett (1976) the Sudbury ores, for example, are not in situ cumulates but deep differentiates intruded later than the main intrusion, an idea which also fits the Notträsk mineralization.

The inner sulfides, poor in Ni, Cu and Co are probably a result of local accumulation in situ. On the other hand, the richer sulfides together with leucocratic veinlets, enrichment of quartz and sericitization of feldspars may indicate that the norite was followed by later magmatic activities – the intrusion of the sulfide vein. The irregular outlines of the veinlets and massive pyrrhotite show that the latest intrusion occurred before the complete solidification of the norite and possibly during the diapiric re-emplacement of the massif.

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